Ji 2020 cSCC — analysis summary

RUN 1 OF 10

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8 patients share one continuous, sample-mixed manifold (Harmony θ=10); detached islands removed.

Gate 2 — module geography → GO (non-branching)

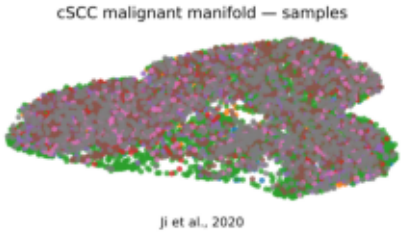
Gast+pSom co-vary (r=+0.13) and both oppose Crest (Gast~Crest=−0.70, pSom~Crest=−0.64): one developmental axis with an inverse terminal Crest gradient (non-branching).

Gate 3 — hub cross-repression → NA

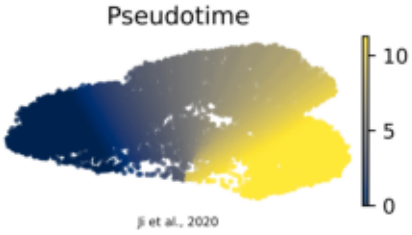
Branching-only test; manifold is non-branching (principal curve: 2 tips, 0 forks) — not applicable.

STEP FIGURES

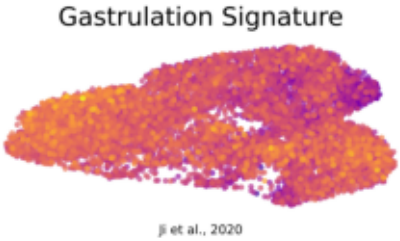
Manifold — samples



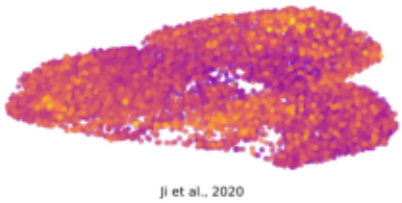
Pseudotime



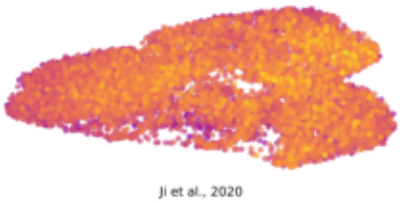
Gastrulation



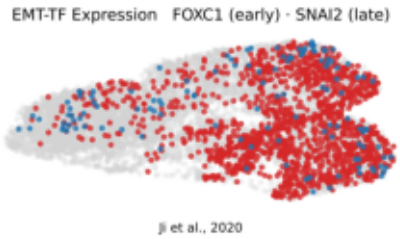
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous Squamous-Cell Carcinoma (Ji et al., 2020) — 10,531 malignant keratinocytes from 8 cSCC patients (published malignant calls; thin samples removed by a data-driven multiplicative-gap threshold). Harmony integration on patient (θ=10), principal-curve trajectory (ElPiGraph) on the whole manifold. Module means: Gast −0.032, pSom +0.042, Crest +0.023 — pSom/Crest globally stronger than Gast.

Verdict.
Non-branching — Linear pSom/Crest. A single developmental axis: a Gast/pSom-high differentiating-keratinocyte body (trajectory root) grading into a neural-crest-high basal/TSK leading edge (terminus). Along pseudotime Gast (ρ=−0.51) and pSom (ρ=−0.44) fall while Crest (ρ=+0.37) rises. The axis is verified separable from proliferation (Gast/pSom↔Crest opposition survives partialling the cell-cycle score). Gate 1 GO, Gate 2 GO, Gate 3 NA. EMT-TFs FOXC1 (early) → SNAI2 (late).

Parameter choices

Malignant cells	10,531 (8/10 patients kept)	**Sample threshold**	multiplicative-gap geomean (drops P3, P5)
Integration	Harmony on patient, θ=10 (mixing-sweep selected)	**Embedding**	PCA 20 PC · neighbours 30 · UMAP spread 1.3
Trajectory	ElPiGraph curve · Nodes=20 · λ=0.3 · μ=100	**Topology**	2 tips, 0 forks · fold-free (self-approach>1)
Root	Gast/pSom-high, Crest-low tip (data-derived)	**Modules**	Gast 76/81 · pSom 88/98 · Crest 101/104 genes
EMT-TF pair	FOXC1 (early) → SNAI2 (late)	**Scope**	whole manifold (holds low-cycling too)